Video Transcript

Video 12: Benefits of Serverless Architecture (03:49)

By now, you have done multiple types of installations and you've worked locally, you've worked with the cloud and you've worked with containers and you've worked with variations of these types of configurations and architectures. If we look at the top left, this is your traditional environment where you're doing the installation locally, where you're doing the traditional 3-tier application. If you look at the next slide over, you can see there that we're doing a lot of the same, but we're doing it now in the cloud, we have additional tiers. Now, you could take a variation of that and run some of those components in the cloud, but now in virtual machines, as you can see there.

Now, we took an even more advanced example and we ran that on top of containers using a container registry. Now, we discussed about how you could take some of the products that are offered in the cloud that have been packaged in a very convenient way in which you transfer some of the responsibility to the cloud provider and you use a managed cloud service, and a managed cloud database is a great idea and something that is very convenient to leverage in many conditions. Now, it might be that you're even greedier and you want even more simplicity, in which case you might want to try a serverless database as the one we just tried.

Now, it might interest you to know that when it comes to the impact on the organization and the amount of people that are involved, the decision making, the coordination that needs to be had when it comes to realizing and using these technologies in a project. As you look there from left to right, you can see that the teams are much smaller, they're much more focused, and they require less coordination since you have less people now, it makes sense that on the left hand side, when organizations were still building their own data centers and they had large teams and they had to solve a lot of the problems that ultimately have been addressed in a much more sophisticated way by the cloud providers, and they have done so all over the world, you have to replicate a lot of that locally. Now, as time went on and we got some better building blocks like virtual machines, the teams became smaller and we used those as templates to be able to move faster. By the time containers were mature enough for us to be able to use, we had a great abstraction there to build on top of.

And as you saw within our course, you were able to span a great deal of databases with minimal effort when it came to the configuration and being able to do an implementation. As you can see there, we can then deploy in seconds; we can live for minutes as we did many times or leave it around for hours. But we have that flexibility, we can do it in demand, we can do it on demand, we can do it programmatically. Now, when it comes to that last tier server list, you can see there that you do not need a large team. There's a lot of configurations that have been hidden from you. There's a lot of convenience that is available there as well. Now, if this is a good fit for what you're trying to achieve, and then you have a pretty small team that could take care of this type of scenario in architecture.

So, with that in mind, I hope you have enjoyed the database section. There's a great deal of richness here. There's a great deal of products that are used the world over by some of the largest organization, and there's a lot here to dig more into. And so, we will leave that up to you.